GLANSIS ORGANISM IMPACT ASSESSMENT

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SPECIES INFORMATION

Scientific Name: *Procambarus clarkii*

Common Name: Red swamp crayfish

IMPACT RESULTS

Environmental: Moderate

Socio-Economic: Moderate

Beneficial: Moderate

Comments: Relative to TM-161 additional information has been added and minor changes to scores have resulted, but the qualitative assessment has not changed as a result.

Model predictions indicated that *Procambarus clarkii* has not fully occupied their suitable climate range, regions with suitable climate for *P. clarkii* are predicted to widen in Europe but contract in North America and Asia (Zhang et al. 2020). Crayfish performance decreased in a long heat wave, suggesting that summer heat waves may have negative effects on *P. clarkii* and that they are unlikely to boost its populations (Carreira et al. 2017).

ENVIRONMENTAL IMPACT

E1) Does the species pose some hazard or threat to the health of native species (e.g., it magnifies toxin levels, is poisonous, a virus, bacteria, parasite, or a vector of one)?

Yes, and it has resulted in the reduction or extinction of one or more native species populations, affects multiple species, or is a reportable disease	6
Yes, but negative consequences have been small (e.g., limited number of infected individuals, limited pathogen transmissibility, mild effects on populations and ecosystems) AND/OR It has significantly affected similar species in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *Many crayfish, including* P. clarkii*, are known to be a source of transmittance of heavy metals among different trophic levels of the food web. Crayfish pass heavy metal contamination on through enriched levels of the metals or pesticides in their organs or tissues, which is then transferred to their consumers (Otero et al. 2003).*
- *The red swamp crayfish harbors numerous flatworm parasites that may be passed on to vertebrates and can carry the crayfish plague fungus (*Aphanomyces astaci*) as a chronic or latent infection (Huner and Barr 1991, Longshaw 2011). It has been implicated in the spread of the fungus to native crayfish in Europe following initial introduction by the signal crayfish (Barbaresi and Gherardi 2000, Mastitsky et al. 2010).North American crayfish species appear to be resistant to most of these diseases (Hunner and Barr 1991).*
- *The white spot syndrome virus, which has caused mass mortalities among shrimp in Europe, can also be carried by* P. clarkii*. Together with its ability to carry the crayfish plague virus, the red swamp crayfish has been characterized within its invaded range as a host to high impact parasites (Mastitsky et al. 2010).*
- Procambarus clarkii *might be a reservoir for* Batrachochytrium dendrobatidis *(Bd), a fungus that causes disease in amphibians (Oficialdegui et al. 2019).*
- *By reducing intermediate odonate predators,* P. clarkii *increases mosquito populations in California (Bucciarelli et al. 2018).*

E2) Does it out-compete native species for available resources (e.g., habitat, food, nutrients, light)?

Yes, and it has resulted in significant adverse effects (e.g., critical reduction, extinction, behavioral changes) on one or more native species populations	6
Yes, and it has caused some noticeable stress to or decline of at least one native species population	1 √
Not significantly	0
Unknown	U

- Procambarus clarkii *is a strong competitor with native crayfish species, such as the white river crayfish (*P. acutus*) or the signal crayfish (*Pacifastacus leniusculus*), and may exclude these species from their shelters (Arrignon et al. 1999, Gherardi and Daniels 2004, Mueller 2007).*
- *P. clarkii initiated 55% more aggressive interactions than P. leniusculus (Meira et al. 2019)*
- *In Italy, the invasion of* Procambarus clarkii *and* Faxonius limosus *were associated with the extinction of* Austropotamobius pallipes *(Manenti et al. 2019)*
- *Faxonius rusticus were significantly more likely to seek shelter when threatened, while P. clarkii were significantly more likely to respond aggressively (O'Shaughnessey and Keller 2019)*
- P. clarkii *won more competitions with Faxonius rusticus for food (O'Shaughnessey and Keller 2019)*
- *Aggression exhibited by the red swamp crayfish has been attributed to reduced breeding success among adult California newts (Gamradt et al. 1997).*

- *Extensive removal of macrophytes by the red swamp crayfish may have led to local extinction of two snails (*Lymnaea peregra, L. stagnalis*) and three plants (*Myriophyllum alterniflorum, Utricularia australis, Ceratophyllym demersum*) in Spain (Montes et al. 1993). Alternatively, direct predation on the snails may have led to the snails' disappearance (Alcorlo et al. 2004).*
- *Herbivorous bird populations (e.g., ducks) have also been severely impacted by the introduction of* P. clarkii *in Spain (Rodríguez et al. 2005).*
- *It has been suggested that populations of the water lily* Nymphaea nouchalii var. caerulea *declined in Lake Naivasha, Kenya as the result of* P. clarkii *herbivory (Hofkin et al. 1991, Lowery and Mendes 1977).*
- *Dragonfly nyph predation efficiency is reduced in the presence of* Procambarus clarkii *(Bucciarelli et al. 2018).*

E3) Does it alter predator-prey relationships?

Yes, and it has resulted in significant adverse effects (e.g., added pressure to threatened/endangered species, significant reduction or extinction of any native species populations, creation of a dead end or any other significant alteration in the food web)	6
Yes, and it has resulted in some noticeable stress to or decline of at least one native species population AND/OR Yes, and it has resulted in some alteration of the food web structure or processes, the effects of which have not been widespread or severe	1 √
Not significantly	0
Unknown	U

This review found few documented impacts to predator-prey relationships in the Great Lakes region, though impacts in other parts of its invaded range are well documented.
Macrophytes:

- Procambarus clarkii *reduces macrophyte diversity through selective macrophyte consumption, which further changes the local environment richness and ecological assemblage (Cronin et al. 2002).*
- *In laboratory studies,* Procambarus clarkii *ate more* Potamogeton nodosus *than* Cherax quadricarinatus *and* Potamonautes perlatus *(Madzivanzira 2022)*
- *Herbivory in red swamp crayfish has been found to have a significant impact on aquatic macrophytes and periphyton (Elser et al. 1994, Lodge 1991, Matthews et al. 1993, Weber and Lodge 1990) and to change the relationships of benthic insects with plants (Hanson et al. 1990, Lodge et al. 1994).*
- *Red Swamp Crayfish are reported to consume Giant Salvinia (Wahl and Diaz 2019)*
- *Preferences (measured as occupancy and loss of macrophyte weight) for* P. clarkii *were* Hydrilla verticillata > Myriophyllum spicatum=Vallisneria denseserrulata = Potamogeton malaianus > Potamogeton crispus *L.* > Ceratophyllum demersum (Yijuan et al. 2022)
- *Results of a field enclosure experiment in Japan suggest that* Procambarus clarkii *may have stronger effects on aquatic macrophytes in shallow lakes and wetlands than common carp (Matsuzaki et al. 2009).*

Macroinvertebrates:

- *Red swamp crayfish juveniles have the potential to significantly reduce local macroinvertebrate diversity through predation (Correia and Anastácio 2008).*
- *Reduction of snails and other grazers through predation may lead to increased periphyton biomass relative to macrophytes. However, prey preference for predatory insects would promote grazer populations and instead decrease periphyton density. (Alcorlo et al. 2004, Evangelista et al. 2019)*
- *Found to prey on dragonfly nymphs in California which caused an increase in mosquitoes in the same area (Bucciarelli et al. 2018)*
- *Procambarus clarkii feeds on many animals, including waterfleas, insects, and other crayfish (Gutiérrez-Yurrita et al. 1998)*
- P. clarkii *may have direct and indirect impacts on bivalve populations by increasing mortality or by reducing their fitness (Meira et al. 2019)*

- *The aggressive nature of* P. clarkii *leaves them more vulnerable to predation (relative to* F. rusticus*) because they rarely flee from threats (O'Shaughnessey and Keller 2019)*

Herps

- *The larval densities of amphibian communities have been shown to be reduced in the presence of* Procambarus clarkii*. (Bélouard et al. 2019)*
- *The disappearance of a newt species in California has been attributed to predation, particularly on eggs and larvae, by* P. clarkii *(Diamond 1996, Gamradt and Kats 1996).*
- Procambarus clarkii *feeds heavily upon the egg masses of* Bufo calamita *and appears to affect the distribution of the species (Cruz et al. 2006)*
- *Field surveys and diet analysis suggest that* Procambarus clarkii *became a significant prey item for the American bullfrog (*Lithobates catesbeianus*), reducing pressure on native species (Bissattini et al. 2018).*
- Procambarus clarkii *represents a major risk to the desert mud turtle (*Kinosternon sonoriense *sonoriense), by preying on hatchling mud turtles (Araya-Donoso et al. 2022).*

Fish

- Procambarus clarkii *feeds upon the eggs and larvae of the endangered razorback sucker (*Xyrauchen texanus*), and given the high density of the crayfish in the species' spawning area, recruitment of the endangered species could be threatened (Mueller et al. 2006)*
- *The presence of crayfish resulted in significantly lower survival rates for adult pupfish (Paulson and Stockwell 2019)*

Energy flow

- *Consumption of detritus by* P.clarkii *significantly increased the rate of leaf litter decomposition (Evangelista et al. 2019)*
- *Consumption of detritus by* P. clarkii *can restructure energy flow (e.g., shortened pathways to top predators, simplified food web structure) through traditional trophic levels in an invaded system (Geiger et al. 2005).*
- *Presence of* P. clarkii *significantly decreased community respiration rate (Evangelista et al. 2019)*

Spread of other invasive species

- *Crayfish invasion can lead to novel secondary dispersal pathways for plants and invertebrates; viable plant and algae propagules were recovered from gull excreta which were similar to those found on the crayfish and likely secondary to gull consumption of the crayfish (Lovas-Kiss et al. 2018)*
- *Ludwigia grandiflora became more fragmented in the presence of a high density of* P. clarkii*, which might increase its dispersal abilities (Thouvenot et al. 2017)*

E4) Has it affected any native populations genetically (e.g., through hybridization, selective pressure, introgression)?

Yes, and it has caused a loss or alteration of genes which may be irreversible or has led to the decline or extinction of one or more native species	6
Yes, some genetic effects have been observed, but consequences have been limited to the individual level AND/OR It has genetically affected the same or similar species in past invasions outside of the Great Lakes	1
Not significantly	0 √
Unknown	U

- *Congeneric* Procambarus acutus *and* Procambarus gracilis *are native to the Great Lakes (NOAA-Great Lakes Water Life 2025)*
- *A study looking for potential congeneric hybridization (of* P. clarkii *with* P. troglodytes*) in North and South Carolina found no evidence of hybrids (Kendrick et al. 2025)*

E5) Does it negatively affect water quality (e.g., increased turbidity or clarity, altered nutrient, oxygen, or other chemical levels/cycles)?

Yes, and it has had a widespread, long-term, or severe negative effect on water quality AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected water quality to some extent, but the alterations and resulting adverse effects have been mild AND/OR It has significantly affected water quality in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *When foraging activities cease,* P. clarkii *constructs new burrows. This burrowing activity increases water turbidity and decreases primary production (Rodríguez et al. 2005).*
- *Foraging and burrowing behavior in* P. clarkii *can lead to changes in water quality and increased nutrient release from sediment, which may induce localized summer cyanobacteria blooms and eutrophic conditions (Angeler et al. 2001, Duarte et al. 1990, Geiger et al. 2005, Nyström et al. 1996, Yamamoto 2010).*
- *Increased turbidity from suspended sediment also can reduce light penetration and primary productivity (Anastácio and Marques 1997, Angeler et al. 2001).*
- *Results of a field enclosure experiment in Japan suggest that* Procambarus clarkii *may have stronger effects on the clarity of shallow lakes and wetlands than common carp (Matsuzaki et al. 2009).*

E6) Does it alter the physical ecosystem in some way (e.g., facilitated erosion/siltation, altered hydrology, altered macrophyte/phytoplankton communities, changes to substrate (physical or chemical))?

Yes, and it has had a widespread, long term, or severe negative effect on the physical ecosystem AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected the physical ecosystem to some extent, but the alterations and resulting adverse effects have been mild AND/OR It has significantly altered physical ecosystems in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *Capable of removing macrophytes from large areas with its cutting feeding behavior (Feminella and Resh 1989, Smart et al. 2002),* P. clarkii *causes major shifts in habitat heterogeneity and reduces habitat availability for many invertebrates, amphibians, and juvenile fishes (summarized in Alcorlo et al. 2004, Nyström 1999).*
- *Increased temperatures cause* Procambarus clarkii *to significantly enhance and increase leaf-litter breakdown and fine particulate organic matter production (Carvalho et al. 2022)*
- *The red swamp crayfish builds its burrows at the water's edge, and collapse is common on soft sediment banks when burrows are abandoned (Barbaresi et al. 2004).*
- *Burrowing activity can affect the nesting ground of demersal fish (Lowery and Mendes 1977).*
- *The presence of* Procambarus clarkii *suppressed seedling emergence from the seed bank in Lake Kasumigauru, Japan (Matsuzaki et al. 2009)*
- *Habitat modification by invasive crayfish such as* P. clarkii *can facilitate its growth through enhanced food accessibility (Nishijima et al. 2017)*

Environmental Impact Total	5
Total Unknowns (U)	0

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate** √
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

SOCIO-ECONOMIC IMPACT

S1) Does the species pose some hazard or threat to human health (e.g., it magnifies toxin levels, is poisonous, a virus, bacteria, parasite, or a vector of one)?

Yes, significant effects on human health have already been observed	6
Yes, but negative consequences have not been widespread, long lasting, or severe AND/OR It has significantly affected human health in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *Through accumulation of heavy metals and cyanobacteria toxins (e.g., microcystin), the red swamp crayfish facilitates biomagnification of these harmful materials and their trophic transfer to humans (Gherardi and Panov 2006).*
- *In parts of the world, undercooked* P. clarkii *may transmit parasites to humans, including lung fluke (*Paragonimus westermani*) and rat lungworm (*Angiostrongylus cantonensis*) (Matthews 2004).Louisiana populations of the red swamp crayfish have also been found to harbor the lung fluke,* P. kellicoti *(Huner and Barr 1991).*
- *In its introduced range in California,* Procambarus clarkii *has been shown to reduce dragonfly nymph density, which in turn resulted in an increase in mosquito larvae density, resulting in loss of ecossytem services to control a disease vector (Lowden 1992).*
- *By reducing intermediate odonate predators,* P. clarkii *increases mosquito populations in California (Bucciarelli et al. 2018).*

S2) Does it cause damage to infrastructure (such as water intakes, pipes, or any other industrial or recreational infrastructure)?

Yes, it is known to cause significant damage	6
Yes, but the costs have been small and are largely reparable or preventable AND/OR It has a history of causing significant infrastructural damage in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *In areas prone to water level fluctuation—such as around dams, levees , or irrigation systems—complex, deep burrows or numerous simple burrows built by red swamp crayfish are especially likely to damage these structures through bank destabilization. Where water levels are more constant (e.g., reservoirs, marshes), burrows tend to be shallow and simple (Correia and Ferreira 1995).*
- *Economic analysis by Kouba et al. (2022) report damage costs of approximately 3 million US$ (2017 value) for* Procambarus clarkii *in Europe. Impacted sectors include agriculture, environment, and fishery.*

S3) Does it negatively affect water quality?

Yes, it has significantly affected water quality, and is costly or difficult to reverse	6
Yes, but the effects are negligible and/or easily reversed AND/OR It has a history of significantly affecting water quality in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *Foraging and burrowing behavior in* P. clarkii *can lead to changes in water quality and increased nutrient release from sediment, which may induce localized summer cyanobacteria blooms and eutrophic conditions (Angeler et al. 2001, Duarte et al. 1990, Geiger et al. 2005, Nyström et al. 1996, Yamamoto 2010).*

S4) Does it harm any markets or economic sectors (e.g., commercial fisheries, aquaculture, agriculture)?

Yes, it has caused significant damage to one or more markets or economic sectors	6
Yes, some damage to markets or sectors has been observed, but negative consequences have been small AND/OR It has a history of harming markets or economic sectors in past invasions outside of the Great Lakes	1 √
Not significantly	0
Unknown	U

- *Internationally,* P. clarkii *has had devastating effects on rice production, preferentially consuming seedlings following rice field flooding and planting, as well as causing water loss and bank collapse due to its burrowing activity (Anastácio et al. 2000, 2005; Correia and Ferreira 1995; Hu et al. 2016).*
- *Predation on fish eggs (e.g., lake trout, Mueller et al. 2006), food competition with commercial fish species, and destruction of fishery nesting and nursing grounds can negatively affect the fishing industry (summarized in Geiger et al. 2005).*
- *In Kenya, the red swamp crayfish has been implicated in the destruction of fishing nets and significant reduction in yield due to damaged fish (Lowery and Mendes 1977).*
- *Wild populations of* Procambarus clarkii *are known to carry white spot syndrome, a crustacean disease that is devastating to the aquaculture industry (Ngamniyom 2020) as well as potentially to commercially-important native crustacean species (Gulf and South Atlantic Regional Panel 2021)*

S5) Does it inhibit recreational activities and/or associated tourism (e.g., through frequent water closures, equipment damage, decline of recreational species)?

Yes, it has caused widespread, frequent, or otherwise expensive inhibition of recreation and tourism	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *The exportation of* Procambarus clarkii *to South Carolina and the potential presence of White Spot Syndrome Virus (WSSV) in this species could infect recreationally important native crustacean species in South Carolina such as the* Callinectes sapidus *and* Litopenaeus setiferus *(Gulf and South Atlantic Regional Panel 2021)*

S6) Does it diminish the perceived aesthetic or natural value of the areas it inhabits?

Yes, the species has received significant attention from the media/public, significantly diminished the natural or cultural character of the area, or significantly reduced the area's value for future generations	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species diminishes aesthetic or natural value of inhabited areas.*

Socio-Economic Impact Total	4
Total Unknowns (U)	0

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate** √
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

BENEFICIAL EFFECT

B1) Does it act as a biological control agent for aquatic weeds or other harmful nonindigenous organisms?

Yes, it has succeeded significantly as a control agent	6
Yes, it has had some success as a control agent, but may be inconsistent or lack a desired level of effectiveness	1
Not significantly	0 √
Unknown	U

- *Red swamp crayfish actively prey on chironomid larvae, a rice pest (Correia and Anastácio 2008).*
- *In Kenya, P. clarkii consumes and competes with the snail vector of schistosomiasis and has thus been used as a biological control agent (Lodge et al. 2005, Allen et al. 2020).*

B2) Is it commercially valuable (e.g., for fisheries, aquaculture, agriculture, bait, ornamental trade)?

Yes, it is economically important to at least one of these industries	6
Yes, but its economic contribution is small	1 √
Not significantly	0
Unknown	U

- *While commercially fished from both native domestic and introduced foreign populations (e.g., Ackefors 1999, Barbaresi and Gherardi 2000), a red swamp crayfish fishery has not been established in the Great Lakes.*
- *MI DNR officers seized 2,000 lbs of live RSC at Sarnia border crossing* https://content.govdelivery.com/accounts/MIDNR/bulletins/200b182

B3) Is it recreationally valuable (e.g., for sport or leisure fishing, as a pet, or for any other personal activity)?

Yes, it is commonly employed recreationally and has some perceived value for local communities and/or tourism	6
Yes, it is sometimes employed recreationally, but adds little value to local communities or tourism	1 √
Not significantly	0
Unknown	U

- *This species' striking red color has led to commercial advertisement as freshwater "lobster" for aquariums (Simon et al. 2005).*
- *The discovery of* P. clarkii *carcasses near popular fishing locations in Michigan suggests that it is being used as bait by anglers (Smith et al. 2018).*
- *Commonly fished for and eaten in its native and introduced ranges.*

B4) Does the species have some medicinal or research value (outside of research geared towards its control)?

Yes, it has significant medicinal or research value	6
It has some medicinal or research value, but is not of high priority OR It is potentially important to medicine or research and is currently being or scheduled to be studied	1 √

Not significantly	0
Unknown	U

- *The red swamp crayfish has been proposed for use as a bioindicator of heavy metals (As, Cd, Cr, Pb, Hg, Ni) and organic compounds (as found in fertilizers and pesticides, for example) due to its propensity to accumulate these environmental contaminants (summarized in Kouba et al. 2010, Richert and Sneddon 2007, Anandkumar et al. 2020).*
- Procambarus clarkii *is used as a test subject for behavioral pharmacological experiments (Gutierrez et al. 2022)*
- *The red swamp crayfish is readily available though the biological supply trade and might be released following classroom or laboratory use (Larson and Olden 2008; Smith et al. 2018).*

B5) Does the species remove toxins or pollutants from the water or otherwise increase water quality?

Yes, it reduces water treatment costs or has a significant positive impact for the health of humans and/or native species	6
Yes, but positive impact for humans or native species is considered negligible	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species improves water quality.*

B6) Does the species have a positive ecological impact outside of biological control (e.g., increases the growth or reproduction rates of other species, fills an important gap in the food web, supports the survival of a species which is threatened, endangered species, or commercially valuable)?

Yes, it significantly contributes to the ecosystem in one or more of these ways	6
Yes, it provides some positive contribution to the ecosystem, but is not vital	1 √
Not significantly	0
Unknown	U

- *In Europe, it has been suggested that high densities of the* Procambarus clarkii *may lead to greater numbers of herons, egrets, and cormorants (Barbaresi and Gherardi 2000, Rodríguez et al. 2005). In Spain,* P. clarkii *provides prey for threatened species (Tablado et al. 2010).*
- *Field surveys and diet analysis suggest that* Procambarus clarkii *became a significant prey item for the American bullfrog (*Lithobates catesbeianus*), reducing pressure on native species (Bissattini et al. 2018).*

Beneficial Effect Total	4
Total Unknowns (U)	0

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate** √
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

Scoring			Qualitative Statement
Score	# U's	**Impact**	
>5	Any	**High**	___ has a high ___ impact in the Great Lakes.
2-5	Any	**Moderate**	___ has a moderate ___ impact in the Great Lakes.
0	0-1	**Low**	There is little or no evidence to support that ___ has significant ___ impacts in the Great Lakes.
1	0		
0	≥2	**Unknown**	Current research on the ___ impact/effect of ___in the Great Lakes is inadequate to support proper assessment.
1	≥1		

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